Yang Zhao

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Education

Northeastern University (NEU) Boston, MA
PhD in Mechanical and Industrial Engineering September 2024 - Present

- Advisor: Prof. Sze Zheng Yong
- GPA: 4.0/4.0

University of Illinois Urbana-Champaign (UIUC) Champaign, IL
MS in Mechanical Engineering August 2022 - May 2024

- Advisor: Prof. Naira Hovakimyan
- GPA: 3.95/4.0

The Hong Kong Polytechnic University (PolyU) Hong Kong, China
BEng (HONORS) in Electronic & Information Engineering September 2018 - May 2022

- Thesis Advisor: Prof. Kenneth K. M. Lam
- GPA: 3.69/4.30 Major GPA: 3.71/4.30 (With First Class Honours)
- Dean's List: 2019 - 2020, 2020 - 2021

Publications

Zhao, Y., Lee, J., Jeon, H. J., & Yong, S. Z. (2026). Learning Safe-by-Design Neural Network Controllers. IEEE Conference on Decision and Control (Under review)

Zhao, Y., Lee, J., Jeon, H. J., & Yong, S. Z. (2026). CAffNet: Hard Constraint-Affine Neural Networks. International Conference on Machine Learning (ICML). arXiv preprint arXiv:2605.24437.

Zhao, Y., Gah, E., & Yong, S. Z. (2025). Tube-Based MPC for Uncertain Sampled-Data Control Systems With Inter-Sample Reachability Analysis. IEEE Control Systems Letters, 9, 3047–3052. https://doi.org/10.1109/LCSYS.2025.3647069

Bansal, A., Yeghiazaryan, M., Yoon, H.-J., Wang, D., Rasul, A. E., Tao, C., **Zhao, Y.**, Zhu, T., So, O., Fan, C., & others. (2025). AirTaxiSim: A Simulator for Autonomous Air Taxis. AIAA AVIATION FORUM AND ASCEND 2025, 3349.

Tao, C., Cheng, S., Wang, F., **Zhao, Y.**, & Hovakimyan, N. (2024). An optimization-based planner with b-spline parameterized continuous-time reference signals. 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 3100–3107.

Bansal, A., **Zhao, Y.**, Zhu, J., Cheng, S., Gu, Y., Yoon, H. J., Kim, H., Hovakimyan, N., & Sha, L. R. (2024). Synergistic perception and control simplex for verifiable safe vertical landing. AIAA Scitech 2024 Forum, 1167.

Research Experience

Intelligent Control and Estimation of Things Lab, NEU July 2024 - Present

Advisor: Prof. Sze Zheng Yong

Graduate Research Assistant

CAffNet: Hard Constraint-Affine Neural Networks

- Proposed CAffNet, a closed-form neural network framework that guarantees hard satisfaction of an arbitrary number of input-dependent affine inequality constraints without requiring full row rank assumptions.
- Proposed a constraint decomposition with a trainable projection module to handle linearly dependent constraints and non-unique projections.
- Evaluated CAffNet with various experiments instantiated with feedforward and transformer architectures, achieving zero constraint violations and better performance.

Learning Safe-by-Design Neural Network Controllers

- Proposed a safe-by-design neural network control framework that integrates trainable CBFs with CAffNet, enabling joint learning of the controller and CBF parameters with constraint satisfaction guaranteed by construction and preserving the expressiveness of control outputs while eliminating the need for a separate safety filtering optimization.
- Introduced a computationally efficient formulation of CAffNet that significantly reduces the time cost of the projection step.
- Demonstrated the effectiveness of our approach through extensive simulations, showing improved safety enforcement and control performance.

Advanced Controls Research Laboratory, UIUC January 2023 - May 2024

Advisor: Prof. Naira Hovakimyan

Graduate Research Assistant

Synergistic Perception and Control Simplex for Verifiable Safe Vertical Landing

- We proposed a verifiable solution for safely landing vertical takeoff and landing (VTOL) vehicles in cluttered environments, with robust collision avoidance guarantees.
- We introduced intercomponent interactions that enhance the system's performance, significantly reducing landing time, while maintaining high reliability of safety guarantees.

An Optimization-Based Planner with B-spline Parameterized Continuous-Time Reference Signals

- We introduced BSPOP that generates continuous-time control signals that can be tracked by the lower-level controllers running at arbitrarily higher frequencies.
- We validated that BSPOP can generate better planning paths compared with the same frequency baseline planner in both simulation and real-world implementation.

Capstone Project, PolyU January - December 2021

Advisor: Prof. Kenneth K. M. Lam

Thesis Student

Low-resolution Face Recognition Algorithms and Noise Effects

- Implemented a face detection and alignment algorithm (Multi-task Cascaded Convolutional Networks (MTCNN)) on facial databases (FERET and SCface)

- Compared the performance of the state-of-the-art low-resolution face recognition (LRFR) method (Identity-Aware LRFR) and a general face recognition algorithm (SphereFace)
- Proposed improved LRFR methods to be robust against noise and achieve higher recognition accuracy in real surveillance systems

Online Summer Research Programme, University of Cambridge July - August 2021
Advisor: Prof. Marwa Mahmoud
Thesis Student

Noise Effects on Low-resolution Face Recognition Algorithms

- Implemented Identity-Aware LRFR on synthetic and real low-resolution images
- Compared the recognition accuracies on those two different kinds of images
- Demonstrated the unsatisfactory recognition performance on real low-resolution images and showed the research gap between the ideal and the real environment
- Evaluated effects of various types of noise (e.g., Gaussian, Poisson, Pepper & Salt) on the performance of Identity-Aware LRFR algorithm

Smart Textiles and Apparel Research Group, PolyU May 2020 - August 2021
Advisor: Prof. Tao Xiaoming
Undergraduate Research Assistant

Paper-Based LED Screen Embedded in Clothing

Remotely Controlled Robot Arm Based on Piezo Sensor

Independent Project Experience

Two-Joint Snake Robot Simulation, Remote June - September 2022
Advisor: Tianyu Wang (Current Ph.D. Student at Georgia Institute of Technology)

- Evaluated the effects of isotropic and anisotropic friction between the swimmer's body and ground in the MuJoCo simulator
- Developed reinforcement learning control algorithms to move the snake in a desired direction on flat ground in the MuJoCo simulator

EIE Microcontroller Contest, PolyU December 2020 - February 2021
Advisor: Dr. Lawrence Chi-Chung CHEUNG

Smart Turnstile System Based on ToF Sensor

- Developed an intelligent turnstile system suitable for both left-handers and right-handers to solve the problem of left-handers unintentionally opening adjacent gates
- Integrated Arduino and ToF sensor to detect and analyze distance data in various turnstile opening gesture cases, enabling the turnstile system to respond correctly to both left- and right-hand gestures

Teaching Experience

ME 270 Design for Manufacturability, UIUC Fall 2023, Spring 2024
Teaching Assistant *Course Instructor: Prof. Tom Golecki*

TAM 212 Introductory Dynamics, UIUC — Spring 2023
Teaching Assistant — *Course Instructor: Prof. Matthew West*

Work Experience

Edison.ai, Tokyo, Japan — June - July 2021
Software Engineer (Remote)

- Independently developed time sequence forecasting models of customer demand used in automated stores to enhance the user experience through adjusting product supply according to specific conditions, including the weather, traffic volume, time, etc.
- Tested the performance of Vector Autoregression (VAR), Long Short-term Memory (LSTM), and Facebook Prophet and demonstrated how to apply them on an air pollution dataset to predict future air quality
- Annotated images with labels (head, hand, and position) by makesense.ai to train human detection models based on YOLO

Awards & Honors

- The Hong Kong Polytechnic University Scholarship 2021/22
- HKSAR Government Scholarship Fund - Talent Development Scholarship 2020/21
- Hong Kong Wanren Funding (2019 Globex Julmester @Peking University)

Extracurricular Activities

Tsinghua i-Space Innovation Camp (1st Prize), Tsinghua University — June - July 2020
Group Member (Remote)

- Made and presented Intelligent Duct Cleaning Robot proposal and business plan

HeartFire Education Service Voluntary Teaching Trip, Guangxi, China — August 2019
Vice Team Leader

- Led a team of 8 volunteers to provide teaching services at the local primary school

Service-Learning Trip, Kyoto, Japan — June 2019
Group Member

- Constructed a treehouse and a wooden bridge to improve tourism and transportation for the local community together with around 30 volunteers

HeartFire, 8th Committee, PolyU — April 2019 - April 2020
Human Resources Manager

- Promoted volunteer teaching activities with other committee members
- Scheduled interviews for around 200 applicants and supervised the recruiting process to select qualified volunteers to teach students in underserved areas

HeartFire Education Service Voluntary Teaching Trip, Hainan, China — January 2019
Team Member

- Provided volunteer teaching services to students at local primary and high schools